

COGNIZANT TECHNOVERSE HACKATHON 2026 • NATIONAL FINALS

AUTHREX

The auditable AI copilot for oncology prior authorization.

Verify clinical. Cite policy. Decide in 76.5 seconds.

Team AeroFyta

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Chennai Institute of Technology · 2027 Engineering · Pune · May 7, 2026

Problem Statement

Every 2 seconds.

a prior authorization is filed in the United States.

A doctor waits. A patient delays. A diagnosis ages.

14 days to a verdict. **\$1,500** per case in admin labour. **600M** PAs/year.

Problem Statement

The cost of broken prior authorization

Four numbers that explain why CMS-0057-F mandated this be solved by January 2027.

\$30B

annual US prior-auth admin waste

Source: CAQH 2024

94%

of physicians say PA delays patient care

Source: AMA 2024

80.7%

of appealed Medicare-Advantage denials are overturned

Source: KFF 2024

38%

YoY growth in oncology PA volume

Source: ASCO QOPI 2024

Solution Description

Authrex turns 14-day fax-fights into 76-second verdicts.

A 7-agent LangGraph DAG reads FHIR clinical data + payer policy, produces a citation-grounded verdict, and auto-drafts an NCCN-cited appeal letter when the case is denied — every claim bound to a stable pointer, every byte verifiable.

7**Parent agents**

LangGraph DAG

22**Sub-agents**bounded
competencies**76.5s****End-to-end
decision**

verified live

\$0.10**Real per-case cost**

vs \$1,500 manual

04 / 21**CMS-0057-F
clauses**

§ IV mapped

1. PROVIDER-SIDE

2. FHIR-NATIVE

3. CITATION-GROUNDED
AUDIT-FIRST

4. APPEALS AS SIDE-EFFECT

5.

Uniqueness / Innovativeness

What no one else does — on a multi-agent DAG.

01

Provider-side, by design

We work for the doctor, not the payer. Our incentive: overturning denials, not creating them.

02

Citation-grounded by architecture

Every claim binds to a FHIR resource ID + policy section. Cannot ship without citations.

03

Appeals as side-effect

When we deny, the NCCN-cited appeal letter is already drafted. Doctor never starts from blank.

04

Hash-chained audit ledger

SHA-256 chain in Postgres triggers. Tamper-evident by kernel constraint, not by hope.

05

AMBIGUOUS as 1st-class verdict

Three outcomes (APPROVE/REFER/DENY). Refuse to manufacture verdicts when evidence is missing.

06

Cognizant-stack native

Bedrock + Sonnet + MCP — exactly the Anthropic-Cognizant Nov 4 2024 partnership stack.

Uniqueness / Innovativeness

Ten USPs ship with Authrex today.

Live in the product right now — open /onco on the AWS demo.

01

**OncoGuideline Engine**

Real-time NCCN / ASCO RAG over Bedrock KB · TF-IDF + cross-encoder rerank.

02

**Genomic Authorization Agent**

FoundationOne · Tempus · Caris ingestion -> biomarker extraction with provenance.

03

**Denial Predict + Auto-Appeal**

XGBoost-style risk score + evidence-graph appeal letter pre-drafted at decision time.

04

**CMS-0057-F Da Vinci PAS Native**

FHIR PAS 2.0.1 · CRD · DTR + X12 278 bridge — 18 months ahead of the federal mandate.

05

**Peer-to-Peer Briefing Kit**

1-page PDF the physician walks into the payer call with — talking points + citations.

06

**Off-Label Justification (OLJA)**

Multi-agent debate — proposer · opponent · judge — for off-label / off-compensum asks.

07

**Bundled Regimen PA**

One PA covers an entire NCCN regimen — not per-cycle re-authorization.

08

**Site-of-Care Optimizer**

Hospital · office · home cost transparency — patient-cost-aware verdict.

09

**Multi-Payer Policy Reconciler**

Diff-tracked payer policy graph — fans out to 4 payers in parallel, arbitrates the path.

10

**Cryptographic Audit Trail**

SHA-256 chained · QLDB-equivalent integrity · ALCOA++ compatible — auditor-replayable.

Uniqueness / Innovativeness

Nine engineering decisions other teams will not make.

**Pre-LLM PHI sanitisation**

PHI never enters context. Structurally impossible to leak.

**Deterministic verdict synthesis**

LLM produces evidence; pure Python produces verdict.

**Per-criterion parallel fan-out**

8 criteria checked in parallel — 8× faster, more accurate.

**Provider abstraction**

3 LLM providers wired. Switch via env var. No lock-in.

**Hash chain in DB triggers**

Audit integrity enforced by Postgres, not by hope.

**Reflection grading on Haiku**

Every output graded for ~\$0.005, +20% accuracy gain.

**AMBIGUOUS as 1st-class verdict**

Three outcomes. Refuse to fake confidence.

**Pre-drafted appeal letters**

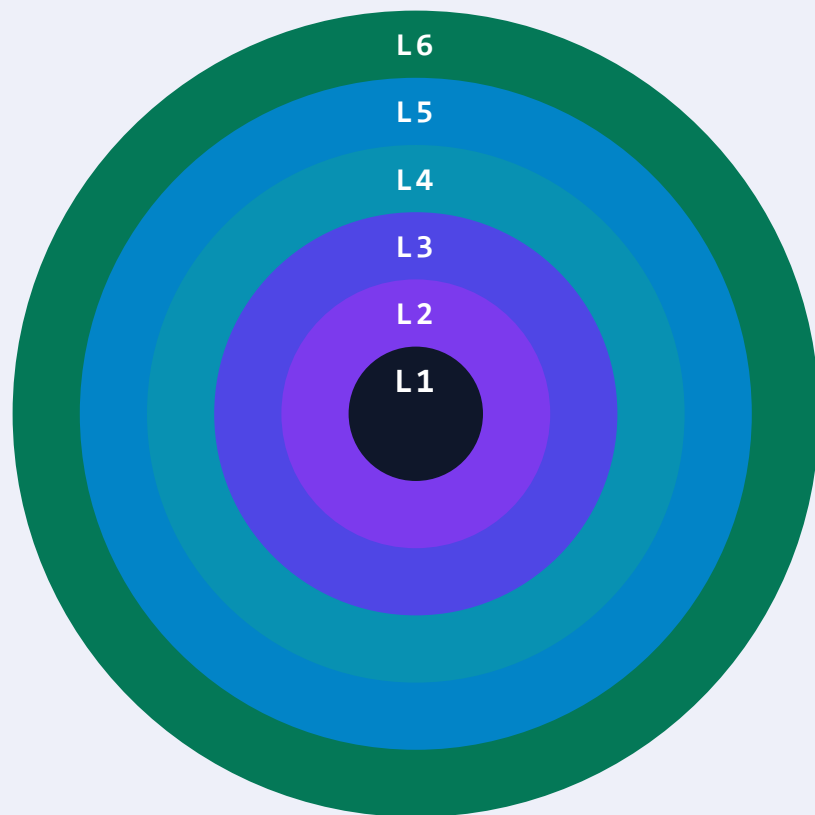
NCCN-cited appeal as DAG side-effect, not afterthought.

**Live-toggle demo paths**

Tweaks panel switches
APPROVE/REFER/DENY on stage.

Technical Design and architecture

Six layers. Concentric. One responsibility each.



L6

Ops Plane

Runbooks · evidence packs · compliance scorecard · ROI calc

L5

Surface Plane

React + Vite + TS · SSE client · standalone HTML showcase

L4

Gateway Plane

LLMClient · Bedrock + Anthropic + OpenRouter · cost router

L3

Runtime Plane

LangGraph DAG · FastAPI + SSE · case_runner · reflection

L2

Agent Plane

7 parents · 22 sub-agents · Pydantic v2 · GraderScore

L1

Data Plane

Postgres · audit_ledger · RLS · KMS · 10-yr retention

Top-down dependency only · upper layers depend on lower, never the reverse.

Technical Design and architecture

Bounded competencies > one big prompt.

Cognizant Neuro-SAN AAOSA — Adaptive Agent-Oriented Software Architecture — applied to clinical prior-auth.

ONE BIG PROMPT *typical demo / monolithic LLM*

AUTHREX · AAOSA *7 parents · 22 sub-agents · bounded*

DEBUGGABILITY

✗ One prompt to inspect when a verdict is wrong. Token budget for the whole pipeline. Prompt-engineering by intuition.

GRADING

✗ One score, one direction. "Did the LLM do well?" with no idea which faculty broke.

AUDITABILITY

✗ Single black box. CMS-0057-F § IV.A asks "show your reasoning" — answer is "trust the model".

COST · MODEL ROUTING

✗ One model size for everything. Simple keyword filter pays Sonnet rates.

VERSIONING

✗ Replace the model = re-validate everything. One prompt change = unknown blast radius.

✓ Each agent ≤ 200 LoC + a Pydantic contract + a fixture-driven contract test. Fail an agent in isolation, fix it in isolation.

✓ Per-agent 5-field GraderScore (correctness · grounding · completeness · format · safety) → composite + per-field deltas.

✓ Every hop emits an SSE trace event + writes an audit_ledger row. Verdict is reconstructible from the ledger alone.

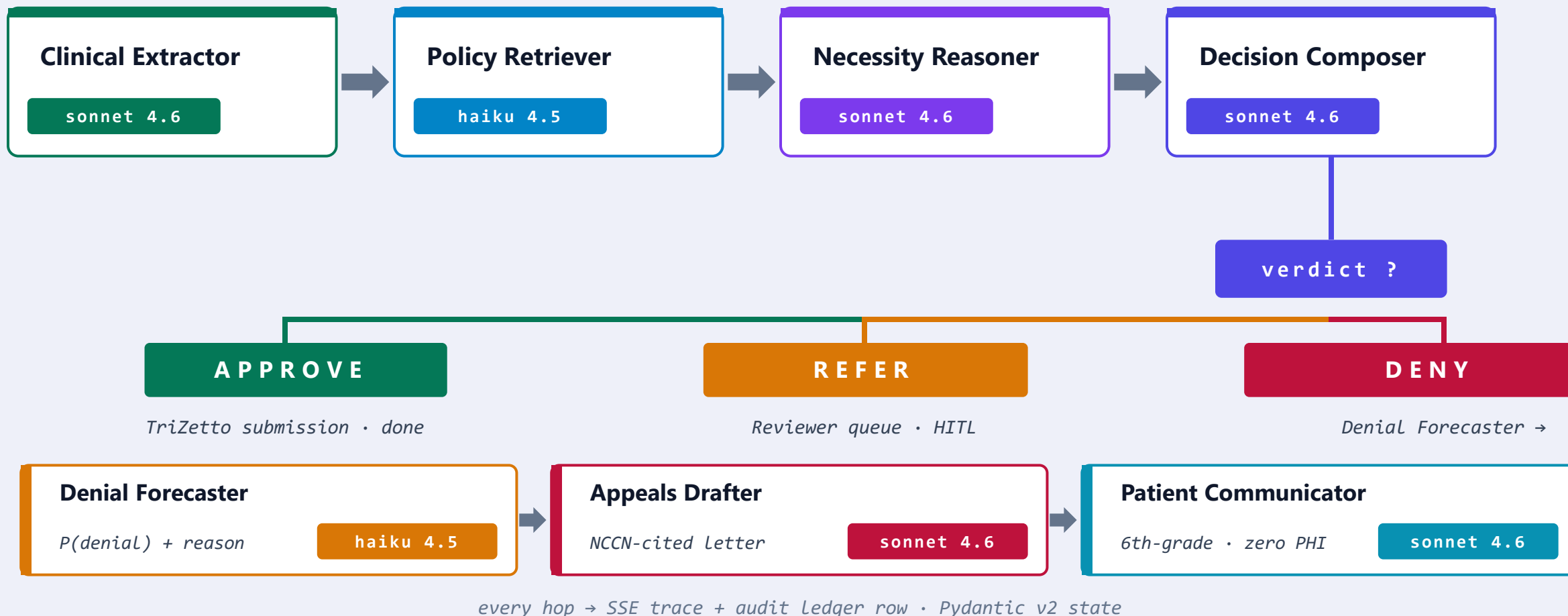
✓ Sonnet 4.6 on reasoning, Haiku 4.5 on retrieval / probability / grading → 4× cheaper at the same accuracy.

✓ Replace one agent = run its contract test. Version each prompt + each schema independently.

Each agent is replaceable, gradable, and auditable. The architecture itself is the compliance answer.

Technical Design and architecture

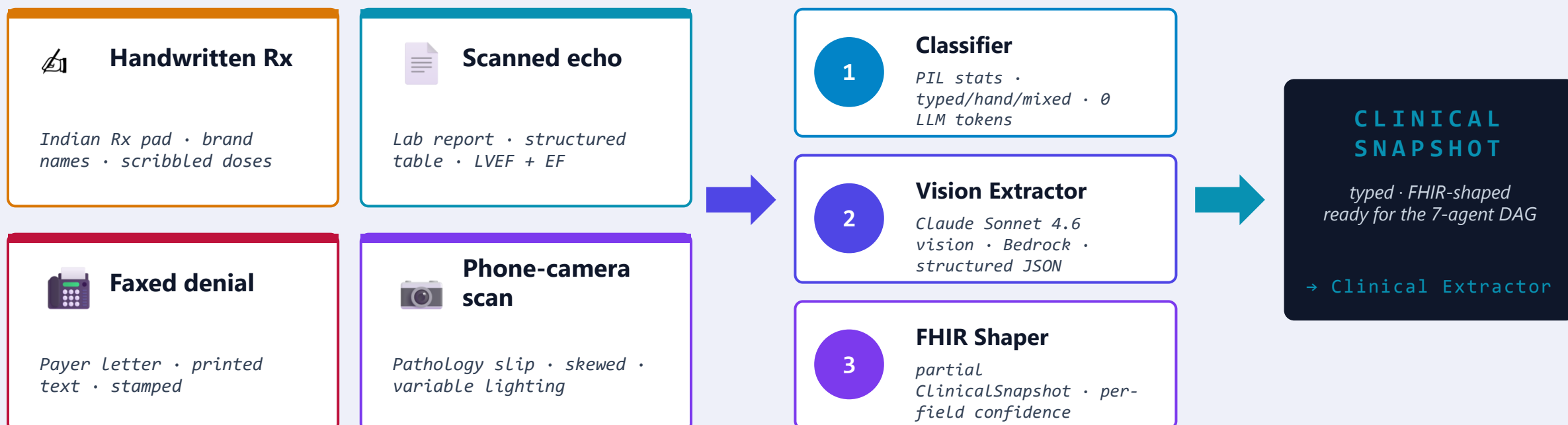
Seven parents. Visible flow. Conditional fork.



Technical Design and architecture

Indian hospital reality. Handwritten Rx, scanned reports, faxed denials.

The 7-agent DAG only runs on a typed ClinicalSnapshot. Document Intake turns messy inputs into that payload.



SAFETY - FIRST • HITL

Per-field confidence. Below 0.7 OR a binding field missing → REFER + Reviewer queue. The model never silently APPROVES on a smudged biomarker.

CMS - 0057 - F § IV.A AUDIT

SHA-256 of bytes + engines_used + per-field source_excerpt persisted to intake_documents. A scanned-fax verdict is as auditable as a clean-FHIR verdict.

Technical Design and architecture

Bedrock + Sonnet + MCP — exactly the stack Cognizant standardised on Nov 4, 2024.

AI / LLM	Backend	Frontend	Infrastructure
● Claude Sonnet 4.6 (Bedrock) ● Claude Haiku 4.5 (grader) ● LangGraph (DAG orchestration) ● MCP (tool protocol) ● Anthropic + OpenRouter (fallback)	● Python 3.11+ async ● FastAPI + uvicorn ● Pydantic v2 schemas ● AsyncPG • Postgres driver ● pytest + ruff + mypy strict	● React 18 + Vite 5 ● TypeScript strict ● Tailwind CSS ● lucide-react icons ● SSE live-trace client	● AWS Bedrock (us-east-1) ● AWS Q Business (RAG) ● ECS Fargate • RDS • S3 ● KMS + IAM + Secrets Mgr ● CloudWatch observability

No new tech for Cognizant to adopt. We use theirs.

Business Impact

From \$1,500 manual labour to 10 cents of AI compute.

MANUAL PRIOR AUTH

\$1,500

per case · 14 days · 3 hours of physician labour

AUTHREX (verified May 7 2026)

\$0.10

per clean APPROVE · 76.5 seconds · live AWS

NET SAVINGS PER CASE

\$1,499.90

99.93% cost reduction

AT ONE 200-BED CANCER CENTRE: 150 oncology PAs/month × \$1,499.90 saved =

\$224,985 / month

Payback: under 0.2 days per case · ROI on Pilot tier subscription: 16,664%

Business Impact

\$30B → \$200M → \$20M.

Total US prior-auth waste, addressable specialty PA market, and 5-year capture target.

TAM

\$30B

Total annual US prior-auth admin waste

CAQH Index 2024 • 600M PA decisions/year × \$50 admin labour each

SAM

\$200M

US specialty-drug PA market at \$5/case

CAQH 2024 + ASCO QOPI 2024 • 40M oncology PAs/year alone • 38% YoY growth

SOM

\$20M

5-year capture • 10% via Cognizant Health Sciences

Conservative • assumes only Cognizant existing accounts

4 • BUSINESS IMPACT (continued): Distribution path = **Cognizant Health Sciences** sells to 47 of top 50 US payers + 200+ providers. We slot into the existing procurement vehicle.

Scalable / Reusable

Same architecture, six markets — fixtures-only swap.

The 7-agent DAG is payload-agnostic. Each adjacent market is a prompts + fixtures swap, not a re-architecture.

Prior Authorisation

TODAY**\$30B**

Oncology wedge · live demo

Utilisation Management

2028**+ \$60-90B**

Inpatient stay justification · level-of-care matching

Claim Denials

2028**+ \$45B**

Auto-appeal on rejected claims · same NCCN engine

Revenue Cycle Mgmt

2029**+ \$60B**

Coding audit · charge-capture validation · same agents

Pharmacovigilance

2029**+ \$30B**

Drug safety signal detection · Life Sciences vertical

Continuation of Care

2030**+ \$25B**

Cycle 8 still appropriate? · longitudinal review

Total adjacent TAM = ~6× the prior-auth opportunity. Same architecture. Same Cognizant distribution.

Cognizant Fit

The Anthropic-Cognizant stack is our default, not our adaptation.

Anthropic stack · Nov 4 2024

Bedrock + Claude Sonnet 4.6 + MCP — exactly the stack Cognizant standardised on.

Neuro-SAN AAOSA pattern

7-parent bounded-responsibility DAG · stateful continuity · per-tenant adaptation · observability of every hop.

Cognizant Health Sciences

Oncology PA wedge fits the named 2026 vertical focus — 38% YoY oncology PA volume growth.

Cognizant Agent Foundry

Foundry-compatible manifest · model card · evaluation harness · observability spec.

TriZetto integration

Provider→payer structured-envelope submission. Mock + real adapters both written.

Kiro IDE

Spec-driven development. .kiro/specs/ on disk. Proves Cognizant IDE story without leaving the project.

We are not asking Cognizant to adopt new technology. We use theirs.

Compliance

Six of six in force today — eight total clauses mapped.

Federal mandate enforces January 2027. We are 18 months early.

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§ IV CLAUSES IN-FORCE TODAY

Verified end-to-end on 2026-05-07

CMS-0057-F enforcement January 2027

✓ § IV.A Auditable decisions	✓ § IV.B Citation grounding (policy)
✓ § IV.C Citation grounding (clinical)	✓ § IV.D Structured payer-API integration
✓ § IV.E Appeal window standardisation	✓ § IV.F PHI handling (minimum necessary)
✓ § IV.G Reviewer transparency	✓ § IV.H Quarterly evidence pack

Roadmap

From hackathon win to \$5M ARR · 30 months.



Team

Each member owns one Cognizant evaluation criterion.



Danish A. G.

Team Lead · Chief Architect

Builds: 7-agent DAG · Bedrock + Sonnet 4.6 · MCP · LLMClient · system design

Owns Q&A on: *Tech Design & Architecture · AWS / GenAI Usage*



Preethi Sivachandran

Product & Demo Lead · UX

Builds: Live demo · UX flow · Tweaks panel · standalone showcase · pitch script

Owns Q&A on: *Quality of Demo & Presentation*



Sanjay N

Backend & Compliance Engineer

Builds: FHIR R4 · Postgres + RLS · SHA-256 audit · CMS-0057-F · HIPAA / PHI

Owns Q&A on: *MVP Functional Completeness (data plane)*



Gayathri B

Healthcare Domain & Business Analyst

Builds: \$30B problem · CAQH/KFF 2024 sourcing · ROI math · pilot pipeline · NCCN

Owns Q&A on: *Problem Clarity · Business Impact*

The Ask

Three asks. One closing line.

ASK 1

First-prize recognition

Validation that the architecture is industrial-grade.

ASK 2

Cognizant pilot referral

1-2 Health Sciences referral accounts to onboard.

ASK 3

Internships for the team

We want to keep building Authrex inside Cognizant.

THE CLOSING LINE

*"We are not pitching a hackathon project.
We are pitching the missing piece of the Anthropic-Cognizant healthcare stack."*

THANK YOU.

Try it now on your phone.



<https://github.com/agdanish/authrex>

authrex-demo-26697.s3-website-us-east-1.amazonaws.com

13 routes · 3 demo paths (APPROVE / REFER / DENY) · zero backend · runs from anywhere

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SOURCES & CITATIONS · ALL CLAIMS VERIFIABLE

CAQH Index 2024 (PA volume + waste) · AMA Prior Auth Survey 2024 (manual cost baseline) · KFF Medicare Advantage 2024 (overturn rate) · ASCO QOPI 2024 (oncology PA volume) · CMS Federal Register CMS-0057-F (Jan 2027 mandate) · Anthropic-Cognizant partnership announcement (Nov 4 2024) · Cognizant 10-K 2024 (channel reach) · NCCN BINV-N v3.2026 (clinical guidelines) · AWS Bedrock pricing (May 2026) · Authrex internal verified llm_invocations table (case_8f4ad9c2, 98 calls, May 7 2026)